

University of the Philippines Los Baños
College of Arts and Sciences
Institute of Computer Science
CMSC 127 – File Processing and Database Systems
2nd Semester AY 2025–2026
Project Specifications
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Introduction

This document outlines the project requirements for CMSC 127, delivered through the laboratory component of the course. It defines the scope, expectations, and deliverables of the final project. Please read this document carefully before requesting clarification from your laboratory instructor.

Rationale

The project is a required component of the laboratory. It essentially reflects all the competencies that the students should have learned from the course.

Project Details

Description

This information system will be a Land Transportation Office (LTO) Information Management System, designed to support the recording and management of drivers, motor vehicles, registrations, and traffic violations in the Philippines. The system aims to simulate a simplified version of real-world LTO operations, emphasizing proper database design, data integrity, and efficient query processing.

Task MSC 12, 21, 22, or 100,

Using the description and details below, design a robust and normalized database and develop a working system using:

- Any programming language (PL) taught in C and
- An RDBMS: MySQL, MariaDB, or PostgreSQL.
- NOTE: Any libraries or tools for DBMS are NOT allowed as well use of frameworks

The system should allow LTO personnel to store, update, and retrieve information related to drivers, vehicles, registrations, and violations accurately and efficiently.

Core Functionality

Driver Management

- Add, update, delete, and search driver records.
- Store driver information such as: Full name, Date of birth, Sex, Address, License number, License type (e.g., Student Permit, Non-Professional, Professional), License status (valid, expired, suspended, revoked), Track license issuance and expiration dates.
- Note: A driver may own multiple vehicles. A driver's name may not be unique.

Vehicle Management

- Add, update, delete, and search vehicle records.
- Store vehicle details such as: Plate number, Engine number, Chassis number, Vehicle type (e.g., motorcycle, private car, public utility vehicle), Make, model, and year, and Color
- Associate each vehicle with a registered owner (driver).

Vehicle Registration Management

- Record vehicle registrations and renewals.
- Track: Registration number, Registration date, Expiration date, Registration status (active, expired, suspended), Maintain a history of registrations per vehicle.

Traffic Violation Management

- Record traffic violations committed by drivers.
- Store violation details such as: Violation type (e.g., speeding, reckless driving), Date and location, Corresponding fine amount, Apprehending officer (optional), Violation status (unpaid, paid, contested), Allow multiple violations per driver and per vehicle.

Reports to be Generated

- Your system should be able to generate SQL-based reports for the following queries:
- View all registered drivers filtered by: License type, License status, Age range, Sex
- View all vehicles owned by a given driver.
- View all vehicles with expired registrations as of a given date.
- View all drivers with expired or suspended licenses.
- View all traffic violations committed by a given driver within a specified date range.
- View the total number of violations per violation type for a given year.
- View all vehicles involved in violations within a given city or region.

Reminders

Milestone Deadlines

- **Submission of ERD:** Submit a clear, complete, and normalized Entity–Relationship Diagram (ERD). **Deadline: Laboratory session on Normalization**
- **ERD Revisions:** Incorporate instructor feedback before proceeding.
- **ERD to Relational Tables:** Convert the finalized ERD into relational tables with appropriate keys and constraints. **Deadline: Laboratory session on SELECT 2**
- **SQL Queries and Database Objects:** Include required queries, views, stored procedures, and/or triggers where appropriate. **Deadline: Laboratory session on TCL**
- **Application Submission and Presentation:** Submit the working application to your lab instructor. **Deadline: First day of Finals Week**
- **Presentation: During Finals Week**

Safety Considerations

- Exercise caution when working during evening or weekend sessions.
- Keep emergency contact information readily available.

Research

- A well-researched system that reflects real-world LTO operations, rules, and constraints will be given higher merit.

Task Assignment

- Divide responsibilities clearly among group members.
- Ensure equitable contribution and adherence to deadlines.

Consultation

- Regularly consult your laboratory instructor for feedback, clarification, and guidance throughout the project.